

Product Carbon Footprint Study Report

Product Name: polypropylene, granulate

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: polypropylene, granulate

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg polypropylene, granulate

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg polypropylene, granulate.

2. System Boundary

polypropylene production, granulate, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of polypropylene, granulate produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **4.324 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

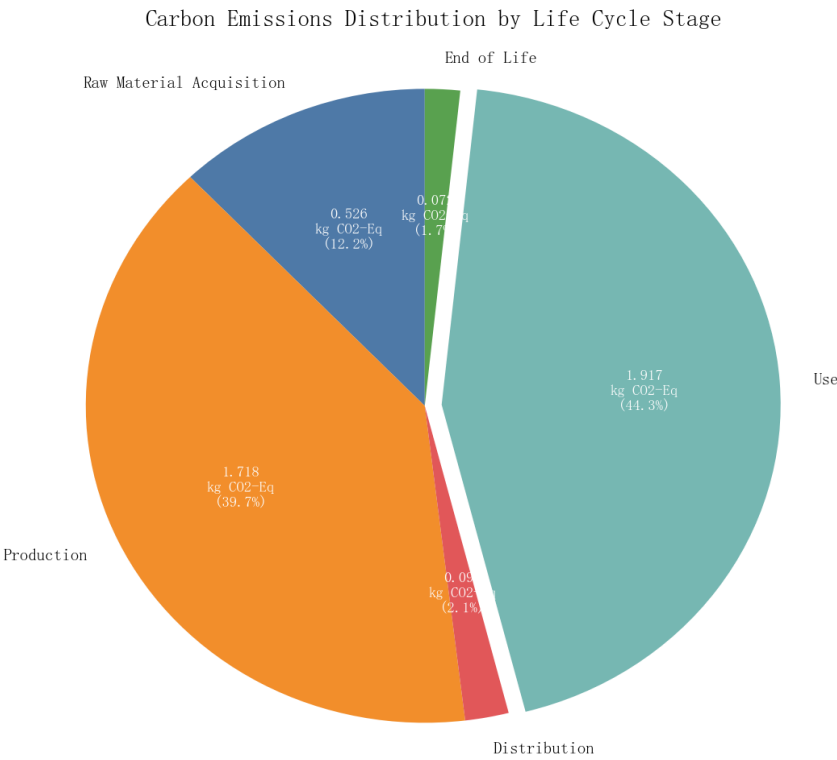


Table 1: Carbon Emissions by Life Cycle Stage

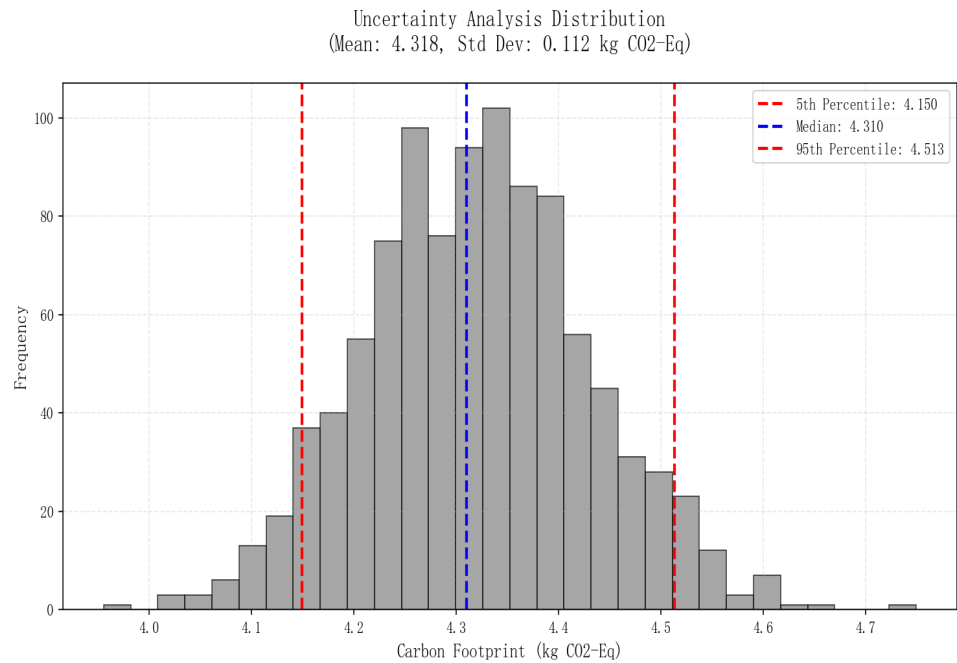
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.526	12.17
Production	1.718	39.74
Distribution	0.090	2.08

Use	1.917	44.34
End of Life	0.072	1.67
Total	4.324	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	4.318
Standard Deviation	0.112
Coefficient of Variation (CV)	2.59%
Median	4.310
5th Percentile	4.150
95th Percentile	4.513

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of polypropylene granulate stands at 4.324 kg CO₂-Eq, representing a moderate emission intensity typical for virgin fossil-based polymers. This aggregated value reflects a full cradle-to-grave assessment and positions the material within the mid-range of commodity plastics when normalized per kilogram.

Lifecycle stage contributions exhibit a clear bimodal pattern, with the use phase (44.3%, 1.917 kg CO₂-Eq) and production phase (39.7%, 1.718 kg CO₂-Eq) together accounting for over 84% of total emissions. Raw material extraction contributes a secondary 12.2%, while distribution and end-of-life stages remain marginal at 2.1% and 1.7%, respectively. This distribution highlights that downstream application energy demands and upstream manufacturing energy inputs constitute the primary emission drivers, rather than logistics or disposal.

2. Statement of Assumptions and Limitations

Key assumptions in this study include the use of average European background data for energy mixes and transport distances, a functional unit of 1 kg of granulate, and an average product lifetime and usage profile representative of typical polymer applications. The analysis further assumes that the use-phase emissions arise predominantly from energy consumption during processing or operation rather than direct material releases.

Limitations stem from reliance on secondary datasets that may not fully capture site-specific production technologies or regional variations in feedstock sourcing, potentially introducing uncertainties of 10–20% in emission factors. Data quality is constrained by temporal gaps in inventory updates and incomplete coverage of certain additives or co-products. System boundary restrictions encompass a standard cradle-to-grave scope but exclude indirect land-use changes, infrastructure construction, and end-of-life scenarios beyond average European waste treatment statistics, thereby omitting potential credits from advanced recycling pathways.

3. Suggestions for Improvement

- Target production-stage emissions by switching to renewable electricity and optimizing reactor efficiency to achieve 15–25% reductions in this dominant phase.
- Address use-phase impacts through material lightweighting or formulation adjustments that lower energy demand during downstream processing or product application.
- Implement supplier engagement programs to source bio-based or recycled propylene feedstock, thereby cutting raw-material contributions.
- Enhance data quality by collecting primary site-specific energy and transport data from key suppliers to reduce reliance on generic averages.
- Explore closed-loop recycling partnerships at end-of-life to shift the small but improvable disposal emissions toward circular credits.

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